PRAJWAL A

Embedded Systems | Embedded Linux | IoT & Electronics
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SUMMARY

ECE graduate (2026) who works from circuit to code: **camera bring-up and on-device vision on Raspberry Pi under Embedded Linux**, a custom **servo-driven PTZ camera**, microcontroller **motor-control** hardware, and 10 months of **production software** experience. Looking for entry-level embedded, firmware, IoT and hardware-validation roles.

PROJECTS

Smart Attendance System – On-Device Face Recognition | *Embedded Linux · Computer Vision*
Raspberry Pi 3/4 · Sony IMX708 NoIR · libcamera / Picamera2 · Python · OpenCV · dlib · Flask · systemd

- Ran the **entire inference pipeline on-device** on a Raspberry Pi under Embedded Linux, with no cloud dependency.
- **Brought up the Pi Camera / IMX708 NoIR module**: fixed camera configuration, image orientation and the purple colour cast typical of NoIR sensors.
- Pipeline: capture, CLAHE lighting normalisation, HOG/CNN face detection, 128-D embedding and matching; CSV/Excel export; live MJPEG preview over **Flask**; runs as a **systemd service** at boot.

Omni Room – AI Smart Classroom: Custom PTZ Camera | *Capstone* Nov 2025 – Mar 2026
Raspberry Pi 4 · Sony IMX sensor · servo motors · 3D-printed enclosure · Python · Ollama

- Designed and assembled a **pan-tilt-zoom camera** from a Pi 4, Sony IMX sensor and **servo motors** in a **3D-printed housing with active cooling**; streamed live video over HTTP to the recognition backend.
- Benchmarked 5 face-recognition approaches before choosing HOG + CNN; system also includes an offline LLM assistant (Ollama) and a 4-role web platform.

RC Car – Microcontroller Drive Control | *Arduino Uno · L298N · HC-05 (UART Bluetooth)*

- Built the full drive chain (Bluetooth input → **Arduino** → **L298N** H-bridge → DC motors) with **GPIO** direction control for forward, reverse and turning.
- Kept motor and logic supplies separate with a common ground; **debugged** wiring/continuity faults, inverted motor direction and **supply dips under motor load**.

PiPilot – Raspberry Pi Voice Assistant | *Raspberry Pi 4 · Python · Linux* In progress

- Speech-to-text / text-to-speech assistant with a **GPIO status LED**, logging, config file and **systemd** autostart; developed headless over **SSH**.

EXPERIENCE

Full Stack Developer Intern | *PulseWriteX Solutions Pvt. Ltd., Bengaluru (Hybrid)* Feb 2026 – Present

- Sole developer of **PulseX** (pulsewritexsolutions.com), a live ordering platform for PCB fabrication, 3D printing and laser cutting that replaced a fully offline quotation and order process.
- Deployed both platforms on **Linux** (Nginx, PM2, CI/CD) and own production support: log monitoring, bug fixing and safe releases with backups and rollback; **100+ production commits**.

TECHNICAL SKILLS

Programming: Embedded C, C, Python, Bash (basic)
Hardware: Raspberry Pi 3/4, Arduino Uno, GPIO, servo & DC motor control, L298N motor driver, sensor/actuator interfacing, power-supply design and debugging
Embedded Linux: Raspberry Pi OS, systemd services, boot-time deployment, libcamera / Picamera2, SSH
Interfaces: UART, I2C, SPI, Bluetooth (HC-05)
Electronics: Analog & digital electronics, circuit debugging with multimeter, continuity and supply troubleshooting
Software & Tools: OpenCV, Flask, Node.js, Git / GitHub, Linux CLI

EDUCATION

B.Tech, Electronics and Communication Engineering | *Dayananda Sagar University, Bengaluru* 2022 – 2026
Relevant coursework: Microcontrollers, Embedded Systems, Analog & Digital Electronics, Signals & Systems, Communication Systems, Control Systems, IoT